MODULE 10

GitHub Copilot Fundamentals for Java Developers

Use GitHub Copilot to accelerate development, write better code, and improve productivity — responsibly.

TEKsystems Global Services

WORKFORCE DEVELOPMENT

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WORKFORCE DEVELOPMENT

GITHUB COPILOT FUNDAMENTALS

1 WHAT IS GITHUB COPILOT?

- AI pair programmer
- Powered by OpenAI codex models
- Trained on public code and best practices

2 KEY BENEFITS

- Boosts productivity
- Reduces boilerplate
- Improves code quality
- Helps learn faster

3 SETUP & INSTALLATION

- Install VS Code or JetBrains extension
- Sign in with GitHub
- Enable Copilot in your IDE

4 HOW IT WORKS

- Understands context from your code
- Suggests entire lines, functions, or blocks
- Keeps learning as you code

5 CORE FEATURES

- Code completions
- Multi-line suggestions
- Function & class generation
- Inline documentation

6 BEST PRACTICES

- Review suggestions
- Understand the code
- Follow coding standards
- Avoid over-reliance

7 USE CASES

- Write boilerplate code
- Generate unit tests
- Refactor and improve code
- Explore new APIs and frameworks

8 PRIVACY & SECURITY

- Your code is private
- Data is not used to train models
- Review license compatibility

MODULE 10 OVERVIEW

AI-assisted development is now part of enterprise software engineering — developers must learn when to trust AI, and when to verify it.

- A developer needs to rapidly scaffold a new customer service. Copilot accelerates development — but every suggestion still needs a human reviewer.

DURATION & LAB

2 Hours Theory

How Copilot works, prompting, generation, review.

45 Min Lab

AI-Assisted Code Generation.

Business Scenario

Scaffold the Northstar CRM customer domain with Copilot.

COPILOT IS

A context-aware AI pair programmer, not an autopilot

YOU DECIDE

Every suggestion is reviewed, tested, and refined by you

THE GOAL

Write better code, faster, without losing engineering judgment

KEY TAKEAWAY AI is a force multiplier — it augments your skills, it doesn't replace your judgment.

LEARNING OBJECTIVES

By the end of this module, you will use GitHub Copilot effectively to write, explain, refactor, and test code.

1. Understand what GitHub Copilot is and how it works.
2. Set up and configure Copilot in your dev environment.
3. Generate context-aware code and complete tasks faster.
4. Write clear prompts for better, more accurate suggestions.
5. Explain and document code with natural language.
6. Refactor and improve code using Copilot.
7. Generate tests and handle edge cases.
8. Review and validate AI-generated code critically.
9. Understand risks, limitations, and ethical use.
10. Apply Copilot to real enterprise use cases.

KEY BENEFITS

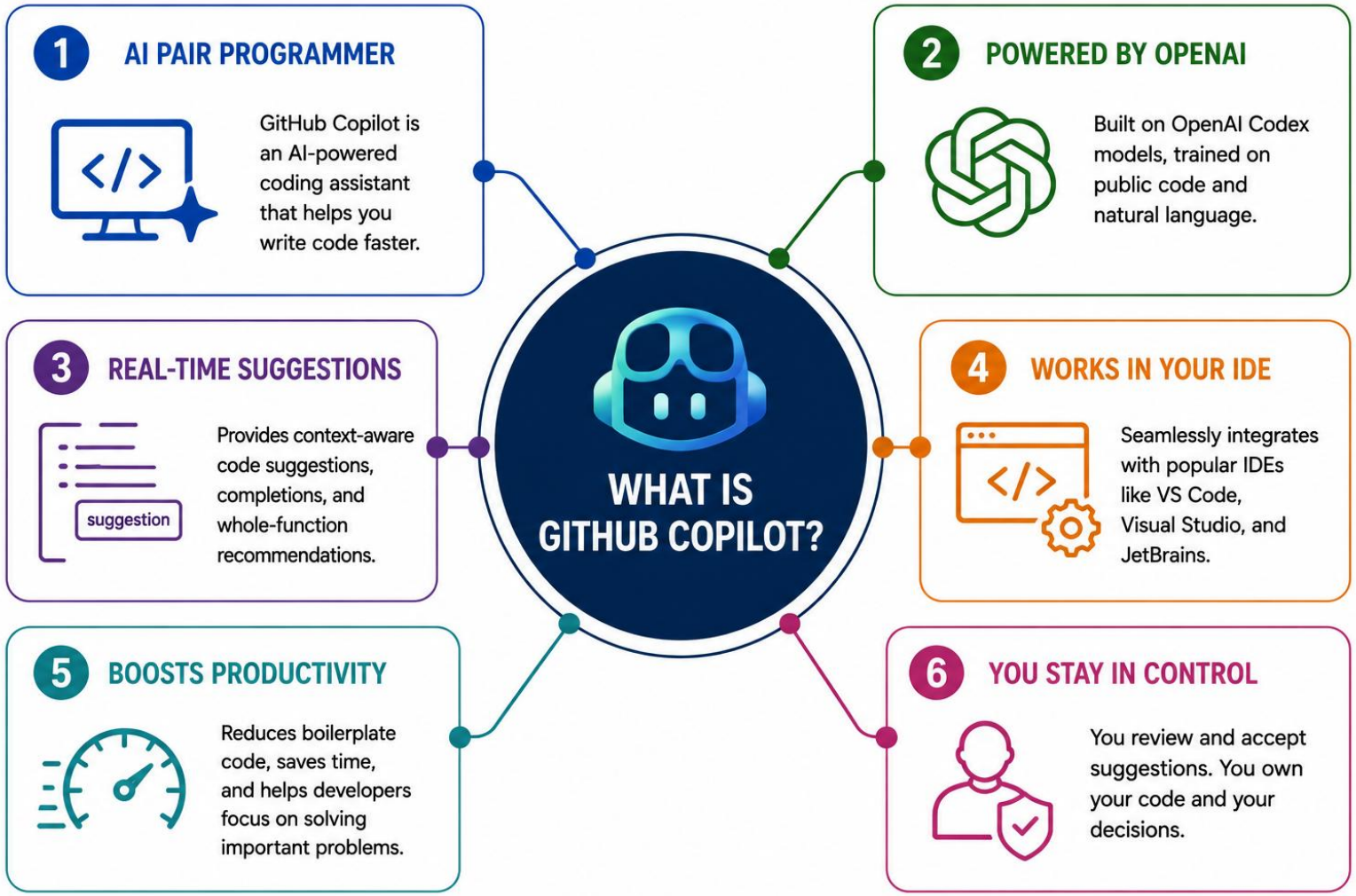
Boost Productivity

Improve Code Quality

Accelerate Learning

Focus on Impact

KEY TAKEAWAY You will be confident using GitHub Copilot to write better code, move faster, and deliver high-quality software.

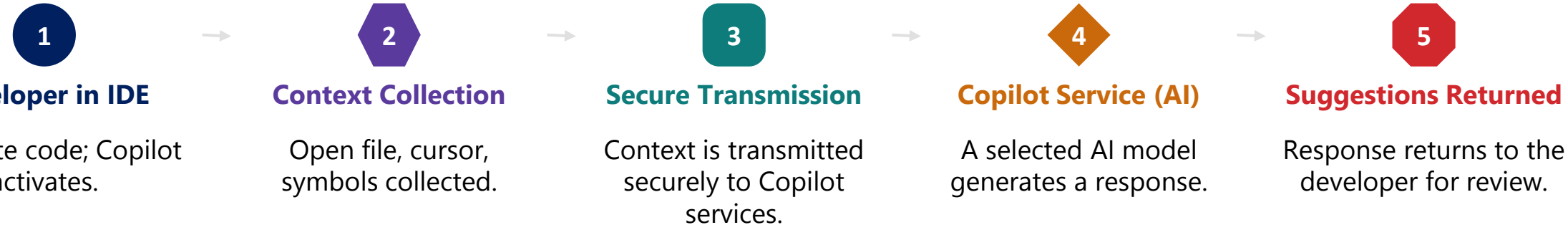


WHAT IS GITHUB COPILOT GitHub Copilot is an AI coding assistant that uses editor and repository context to generate suggestions, explanations, and code-related responses.

It uses large language models selected and operated through GitHub's service, and adapts its suggestions to the context available in your editor, files, repository, instructions, and conversation — not by retraining on your code.

COPILOT ARCHITECTURE OVERVIEW

Copilot integrates into your IDE, securely sends context to GitHub's AI service, and returns intelligent suggestions in real time.



KEY COMPONENTS

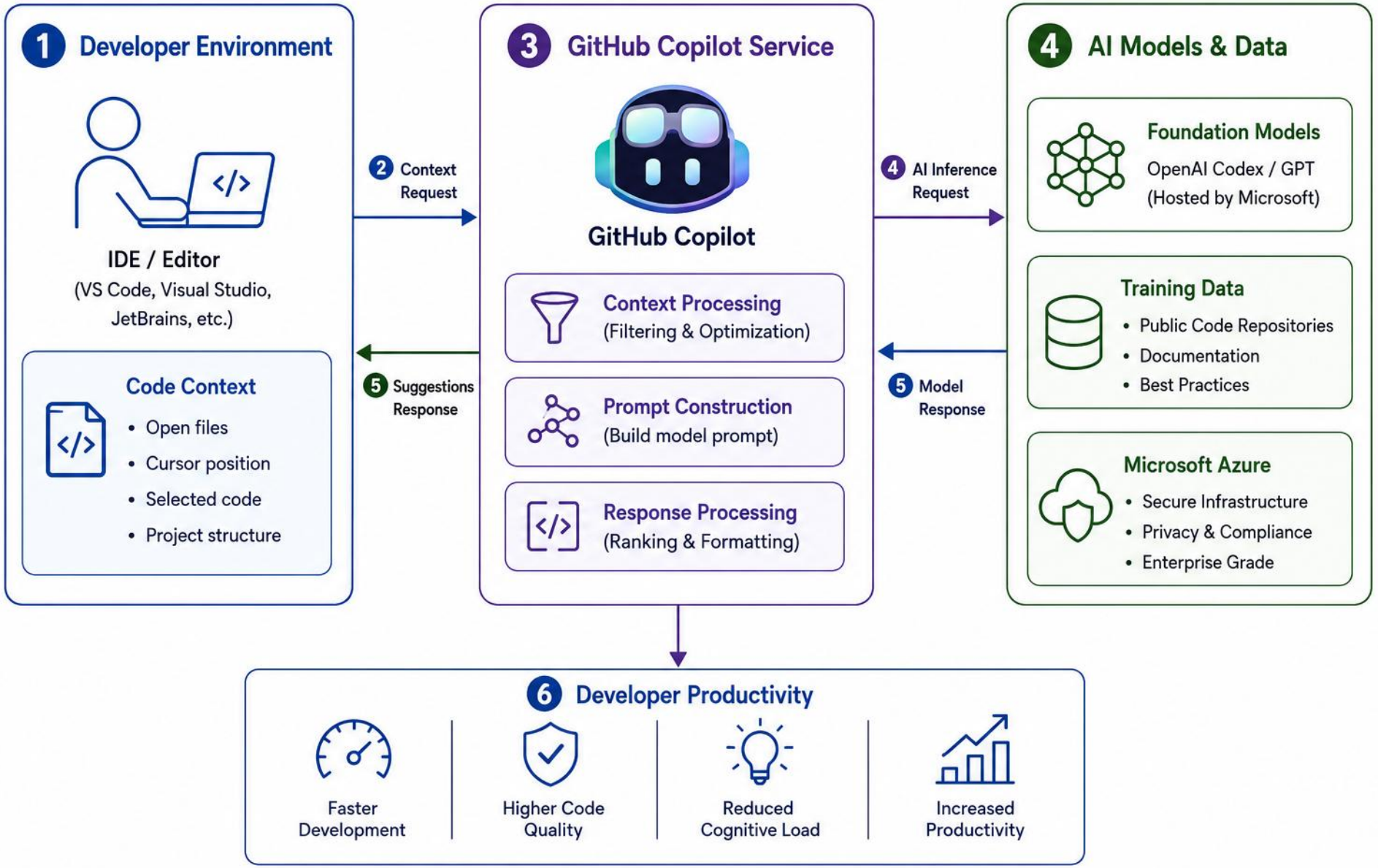
- IDE / Editor — extends your IDE, captures context.
- Context Engine — selects relevant code, filters sensitive data.
- Secure Gateway — encrypts and transmits data.
- AI Service — a selected model generates a response.

SECURITY & PRIVACY

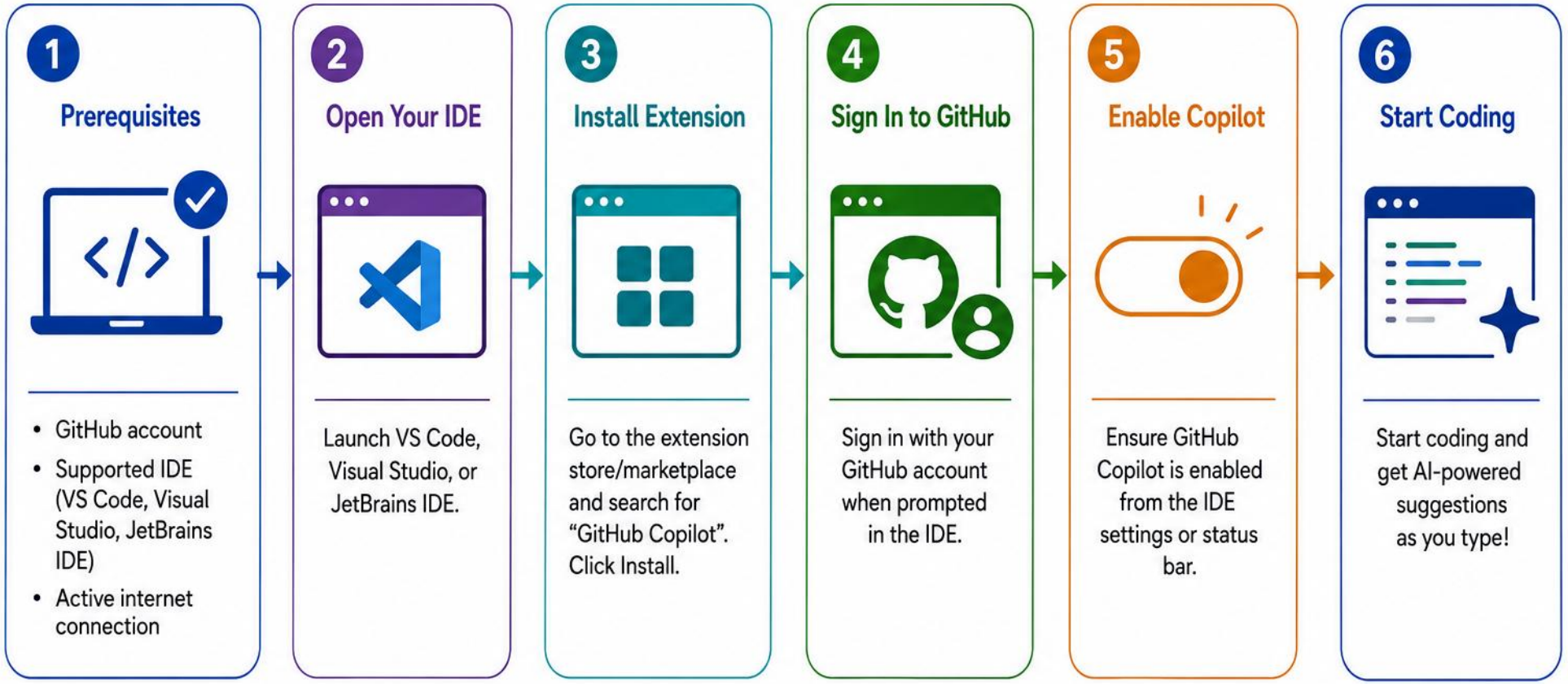
- Data handling depends on your Copilot plan, org policy, settings, and current product terms.
- Follow your organization's Copilot policy at all times.
- Never place secrets, credentials, or regulated data in prompts.

KEY TAKEAWAY Copilot combines large language models, secure cloud infrastructure, and deep IDE integration to deliver real-time assistance.

Copilot Architecture



Installing GitHub Copilot



Note: You need an active GitHub Copilot subscription to use the service.

COPILOT IN INTELLIJ IDEA

Copilot integrates seamlessly with IntelliJ IDEA to help you write, understand, and improve code with AI-powered suggestions.

SETUP IN 3 STEPS

- Go to File > Settings > Plugins > Marketplace, search 'GitHub Copilot', and install.
- Click the Copilot icon in the toolbar and sign in with your GitHub account.
- Authorize Copilot when prompted, then open a file and start typing.

GITHUB COPILOT CHAT

```
> Explain this method.

This method searches for a user in the
given list whose ID matches the provided
id, using Java Streams, and returns an
Optional<User>. Would you like null checks?
```

TIPS FOR BEST RESULTS

Write Clear Comments

Guide suggestions with intent.

Break Into Small Steps

Complex tasks, one step at a time.

Review Carefully

Before accepting any suggestion.

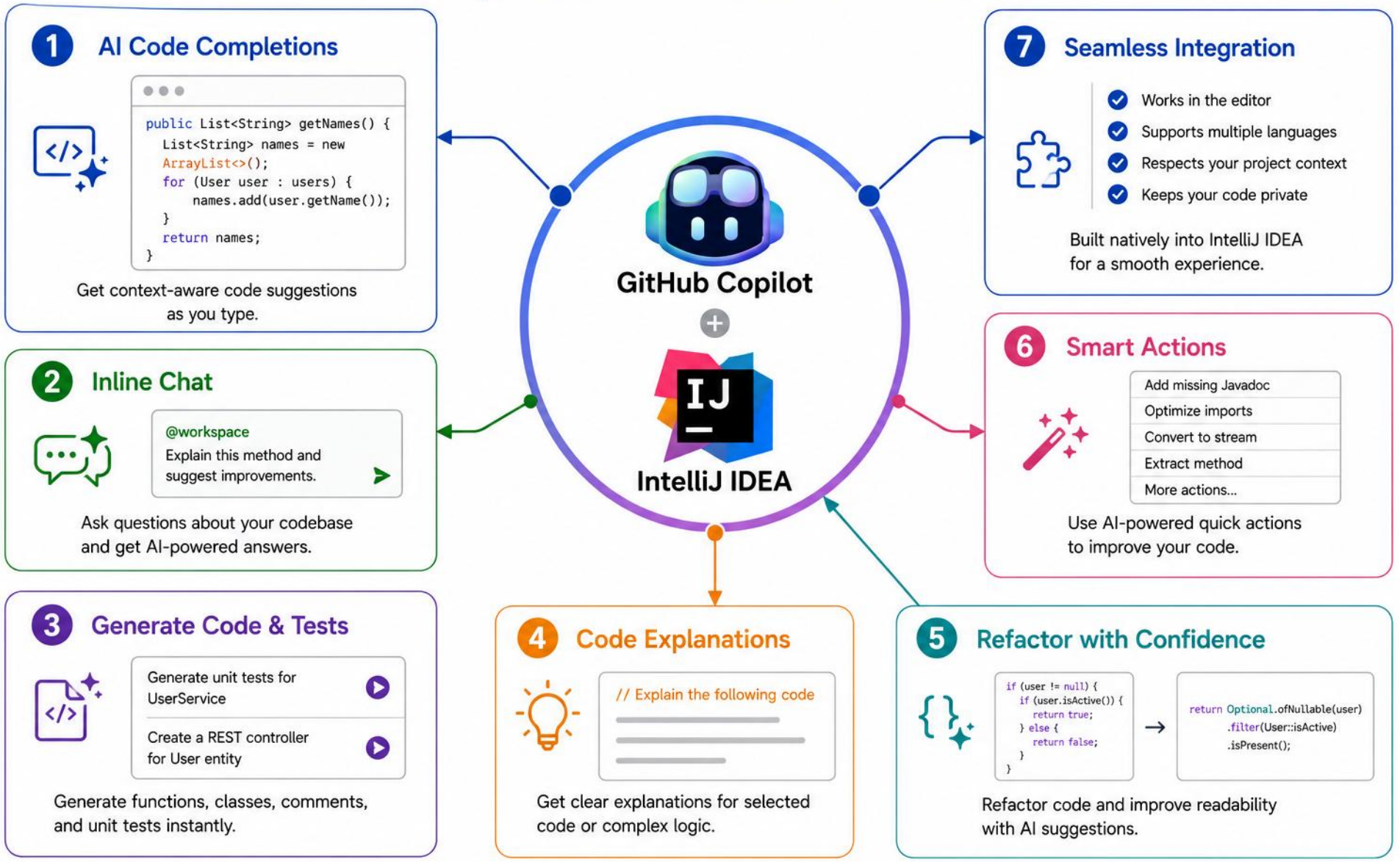
Use Copilot Chat

Learn, refactor, explore alternatives.

- Keyboard shortcuts: Tab to accept a suggestion, Esc to dismiss, Alt+] / Alt+[to cycle through alternative suggestions.

KEY TAKEAWAY Copilot in IntelliJ IDEA acts as your AI pair programmer — understanding context, suggesting code, helping you build faster.

Copilot in IntelliJ IDEA



PROMPT ENGINEERING FUNDAMENTALS

A good prompt helps Copilot understand your intent, use the right context, and generate accurate, high-quality code.

1 **Be Clear**
State what you want.

→

2 **Provide Context**
Share relevant details.

→

3 **Give Structure**
Format, constraints, edge cases.

→

4 **Iterate & Refine**
Improve based on output.

→

5 **Review & Validate**
Test against acceptance criteria.

TASK	WEAK PROMPT (VAGUE)	STRONG PROMPT (EFFECTIVE)
`Sorting`	"Sort a list."	"Sort a List<Employee> by salary, descending, using Comparator."
`API Endpoint`	"Create an API to get users."	"Create a Spring Boot GET /users/{id} returning JSON, 404 if not found."
`Bug Fixing`	"Fix this error."	"NullPointerException in UserService line 45 — user can be null; fix safely."

- Prompt improvement workflow: Write Prompt → Get Response → Review Output → Refine Prompt → Better Output (repeat as needed).

KEY TAKEAWAY Great prompts = goal + context + constraints + expected output + edge cases + acceptance criteria.

PROMPT ENGINEERING IN PRACTICE: WEAK VS STRONG

The same request, two prompts — see the difference in the Java Copilot actually generates.

EXAMPLE: SORTING EMPLOYEES BY SALARY

```
// WEAK PROMPT: "Sort a list."
List<Employee> sorted = list;
// STRONG PROMPT: "Sort a List<Employee> by salary, descending, using Comparator."
List<Employee> sorted = employees.stream()
        .sorted(Comparator.comparing(Employee::getSalary).reversed())
        .collect(Collectors.toList());
```

KEY TAKEAWAY Specific prompts name the type, field, and method — vague prompts force you to fix the output by hand.

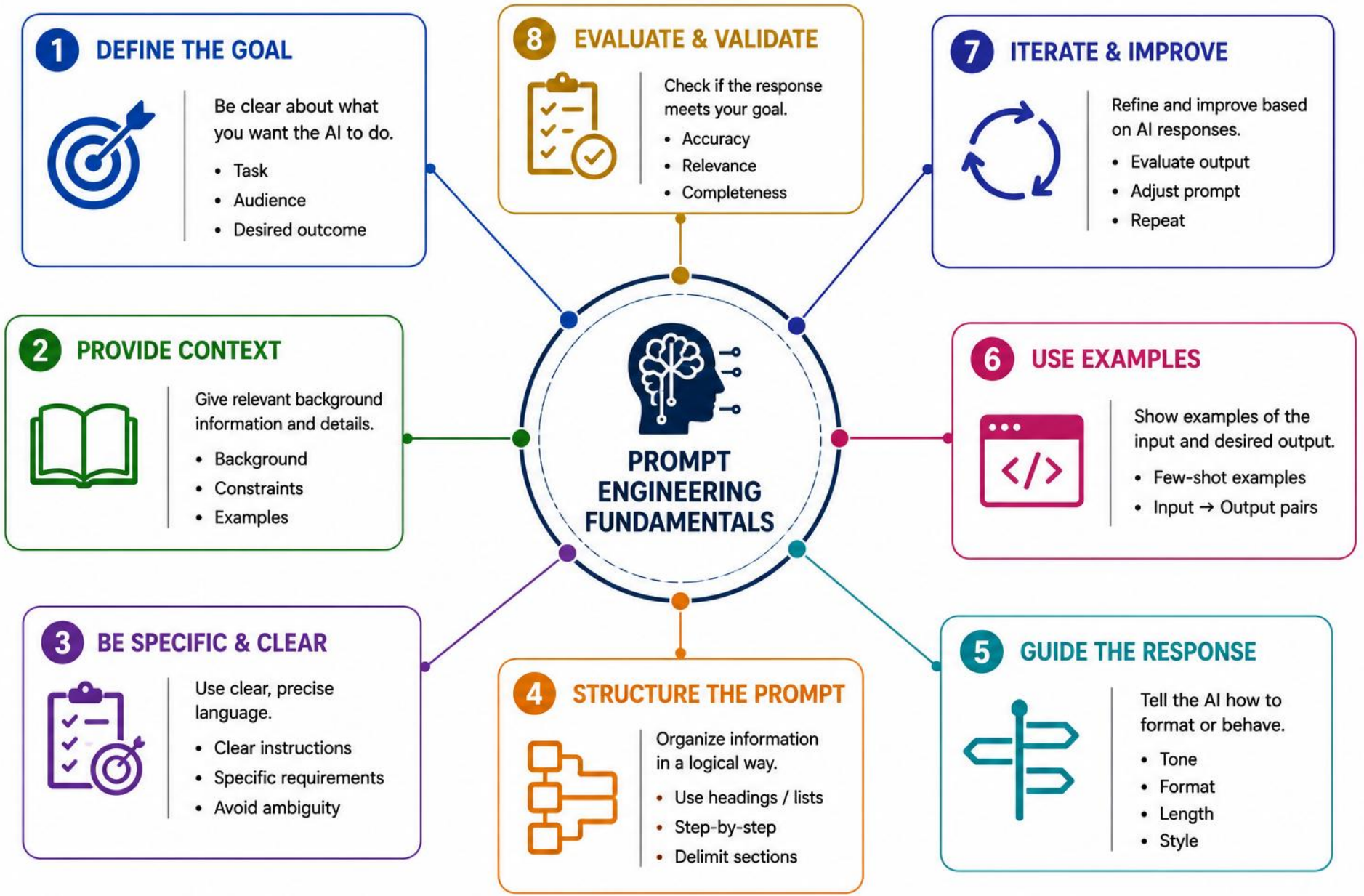
TRY IT YOURSELF — EXERCISE 1: WEAK VS STRONG PROMPTS

Reinforces: the weak-prompt vs strong-prompt table you just saw.

STEP	WHAT TO DO
Goal	Contrast a vague Copilot prompt with a strong Northstar-scoped one
File	lab10-prelab-prompts.md (in examples/module-10-exercises/notes/)
Weak prompt	"Write a customer class." — invites wrong package, JDK APIs, invented annotations
Strong prompt	Java 21 record for CUS-1001 Amina Khan ACTIVE; id, fullName, status; no Spring, no JPA
Constraints	List three: JDK version, domain fixtures, and the no-framework boundary
Reference	exercises/exercise-01-weak-vs-strong-prompts.md

```
$ cat notes/lab10-prelab-prompts.md
Weak:   "Write a customer class."  // wrong package, JDK APIs, phantom annotations
Strong: Java 21 record, Northstar CRM,
        CUS-1001 Amina Khan, status ACTIVE  // id, fullName, status only
Constraints: JDK 21, domain fixtures, no Spring/JPA
```

TRY IT NOW Complete Exercise 1 before continuing — see exercises/exercise-01-weak-vs-strong-prompts.md.



GENERATING DOMAIN CLASSES AND DTOS

Copilot can scaffold a domain class and its DTO from a clear prompt — you review field types, validation, and where the boundary belongs.

HOW TO GENERATE A CLASS OR DTO

- Write a clear comment describing the class you want to generate.
- Begin typing the class declaration — Copilot suggests the rest.
- Press Tab (or Enter) to accept the suggested code.
- Review the generated code and refine names, fields, or logic.

EXAMPLE PROMPT

```
// Create a Java class Customer with customerId
// (Long), fullName (String), email (String),
// phone (String), status (CustomerStatus),
// createdAt (LocalDateTime). Include getters,
// setters, constructors, and toString().
public class Customer {
```

DTO ANNOTATIONS AND BEST PRACTICES

Once Copilot drafts the class, apply these annotation patterns to validate, build, and serialize it correctly.

DTO ANNOTATIONS — WHAT GOES WHERE

Bean Validation

@NotNull, @NotBlank, @Size.

Lombok

@Getter, @Builder, @RequiredArgsConstructor.

Jackson

@JsonInclude, @JsonProperty.

Avoid @Data by Default

Also sets equals/hashCode/toString.

- DTO PROMPT: "Create a DTO for creating a customer with validation annotations for fullName and email." DIFFERENCES TO REVIEW: a DTO carries no business logic — keep it focused on only the fields the API needs, map to/from Customer explicitly, and keep request/response DTOs separate from the entity.

KEY TAKEAWAY Copilot drafts a first pass of a class or DTO — review, test, and align it with your standards before production use.

DTO ANNOTATIONS IN PRACTICE

The annotation patterns from the previous slide, applied to a real request DTO.

GENERATED DTO WITH VALIDATION (CreateCustomerRequest.java)

```
public class CreateCustomerRequest {

    @NotBlank(message = "fullName is required")
    @Size(max = 120)
    private String fullName;

    @NotBlank(message = "email is required")
    @Email
    private String email;
}
```

KEY TAKEAWAY Validation lives on the DTO, not the entity — Copilot drafts the annotations, you confirm the messages and constraints.

TRY IT YOURSELF — EXERCISE 2: CUSTOMER SKETCH FOR AMINA

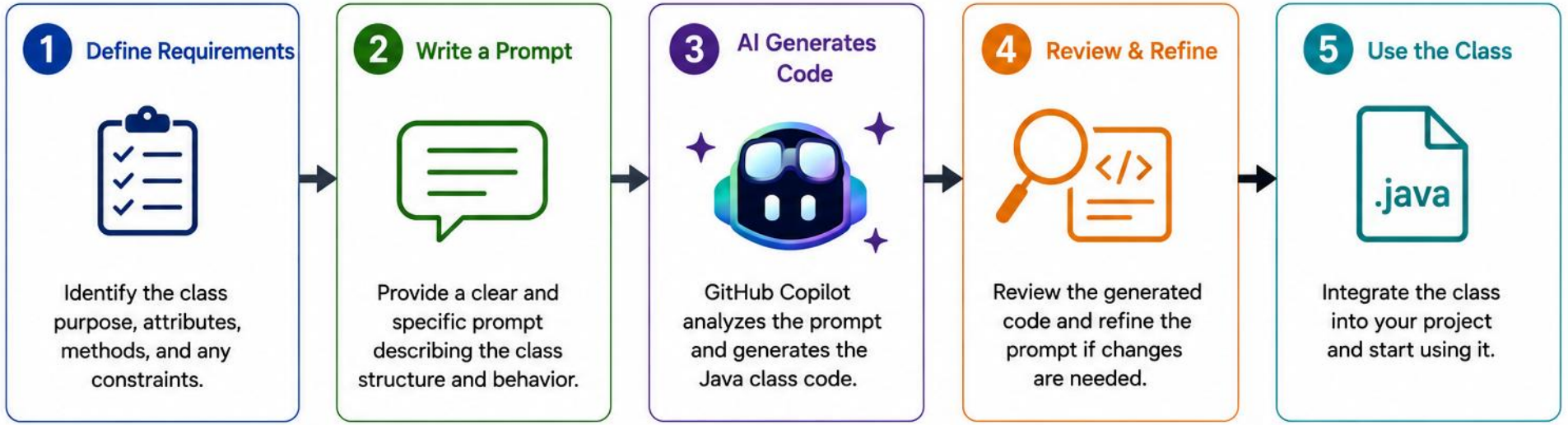
Reinforces: the Customer class fields Copilot just generated for you.

STEP	WHAT TO DO
Goal	Hand-sketch fields for CUS-1001 before asking Copilot to generate any code
File	customer-sketch-notes.md (in examples/module-10-exercises/notes/)
Amina row	CUS-1001, Amina Khan, status ACTIVE — customerId, fullName, status
Ravi row	CUS-1002, Ravi Singh, status PROSPECT — same shape, different values
Correlation note	lab-request-001 belongs in logs/headers later, not as a Customer field
Reference	exercises/exercise-02-customer-sketch.md

```
$ cat notes/customer-sketch-notes.md
customerId   fullName     status
CUS-1001     Amina Khan   ACTIVE
CUS-1002     Ravi Singh   PROSPECT
lab-request-001  // logs/headers only — not a Customer field
```

TRY IT NOW Complete Exercise 2 before continuing — see exercises/exercise-02-customer-sketch.md.

Generating Java Classes



Example Prompt

```
Create a Java class named Student
with the following:
- Private fields: id (int), name (String),
  email (String)
- A constructor to initialize all fields
- Getters and setters for all fields
- A method toString() to display
  student details
```

GitHub Copilot

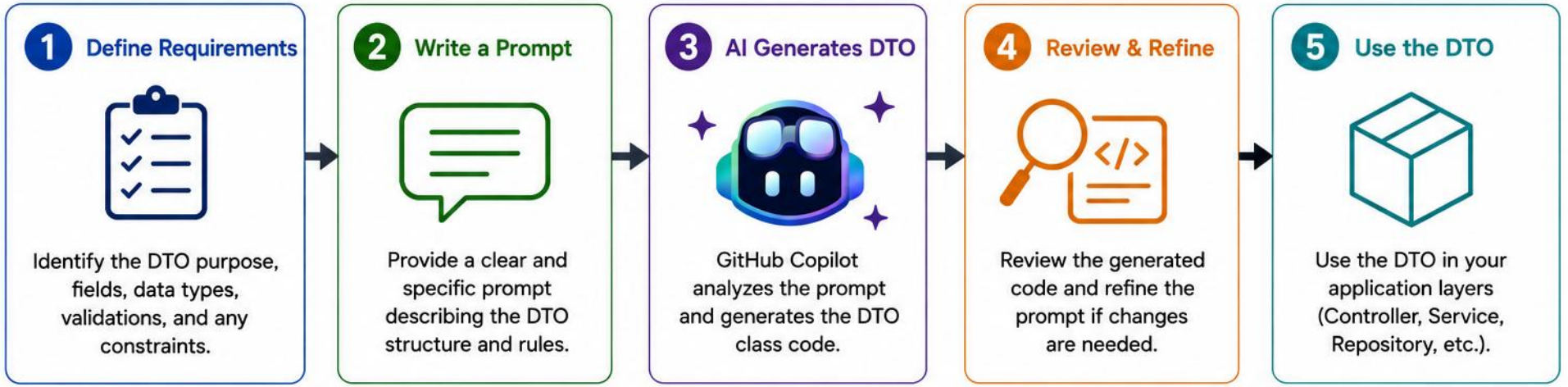
Generated Java Class

```java
public class Student {
    private int id;
    private String name;
    private String email;

    public Student(int id, String name, String email) {
        this.id = id;
        this.name = name;
        this.email = email;
    }
    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }

    @Override
    public String toString() {
        return "Student{id=" + id + ", name='" + name +
               "', email='" + email + "'}";
    }
}
```

Generating DTOs



Example Prompt

Generate a Java DTO class named UserDto with the following:

- Fields: id (Long), name (String), email (String), age (Integer), active (Boolean)
- Private fields
- No-args and all-args constructors
- Getters and setters
- toString() method

GitHub Copilot

Generated DTO (UserDto.java)

```java
public class UserDto {
    private Long id;
    private String name;
    private String email;
    private Integer age;
    private Boolean active;

    public UserDto() {}

    public UserDto(Long id, String name, String email,
                   Integer age, Boolean active) {
        this.id = id;
        this.name = name;
        this.email = email;
        this.age = age;
        this.active = active;
    }
    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }
    public Integer getAge() { return age; }
    public void setAge(Integer age) { this.age = age; }
    public Boolean getActive() { return active; }
    public void setActive(Boolean active) { this.active = active; }

    @Override
    public String toString() {
        return "UserDto{" + "id=" + id + ", name='" + name + '\'' +
                ", email='" + email + '\'' + ", age=" + age +
                ", active=" + active + '}';
    }
}
```

GENERATING SERVICE LOGIC

Service classes hold business logic — validation, search/filter/update rules, and error handling — that Copilot can draft and you refine.

HOW TO GENERATE A SERVICE

- Describe the service's responsibilities, methods, and key dependencies.
- Begin typing the class name or key methods — Copilot suggests the rest.
- Focus on business logic — Copilot fills in method bodies and structure.
- Review generated code and refine error handling and validations.

INDUSTRY PREVIEW — SPRING BOOT ANNOTATIONS

ANNOTATION	PURPOSE
`@Service`	Marks a business-layer component.
`@Repository`	Marks a data-access layer component.
`@Transactional`	Runs the method in a transaction.
`@Valid`	Validates method parameters.
`@RequiredArgsConstructor`	Generates constructor (Lombok).

SERVICE LOGIC IN PRACTICE

See how the CustomerService for Northstar CRM puts these prompts and annotations to work in plain Java.

CUSTOMERSERVICE — PLAIN JAVA

Service Prompt

Add, findById, search, updateStatus.

Business Rules & Errors

Validate input; throw clear exceptions.

Testability

Inject the list; unit test with JUnit.

KEY TAKEAWAY Copilot can draft service logic fast — you still own validation, error handling, and tests before it ships.

NORTHSTAR CUSTOMERSERVICE — PLAIN JAVA

No Spring, no JPA — the in-memory service Lab 10 asks you to scaffold and review.

CustomerService.java (excerpt)

```
public class CustomerService {
    private final List<Customer> customers;

    public Customer findById(String customerId) {
        return customers.stream()
                .filter(c -> c.getCustomerId().equals(customerId))
                .findFirst()
                .orElseThrow(() -> new NoSuchElementException(customerId));
    }

    public void updateStatus(String customerId, CustomerStatus status) {
        findById(customerId).setStatus(status);
    }
}
```

KEY TAKEAWAY Constructor-injecting the List<Customer> keeps the service unit-testable without Spring or a real database.

TRY IT YOURSELF — EXERCISE 3: PHANTOM ANNOTATION HUNT

Reinforces: the Spring Boot annotations you just saw marked Industry Preview.

STEP	WHAT TO DO
Goal	Flag Copilot-style annotations that do not belong in a plain Java prep sketch
File	phantom-annotation-notes.md (in examples/module-10-exercises/notes/)
Defer	@Entity/@Table (JPA), @Service/@Autowired (Spring) — not Lab 10 scope yet
Reject rule	Reject any import you cannot name from JDK 21 or an agreed Maven dependency
Fixture check	CUS-1002 Ravi hard-coded as ACTIVE is wrong — correct status is PROSPECT
Reference	exercises/exercise-03-phantom-annotation-hunt.md

```
$ cat notes/phantom-annotation-notes.md
@Entity / @Table  // JPA only — defer, not Lab 10 scope
@Service / @Autowired  // Spring — defer, hosting labs later
@NotNull (Jakarta)  // name it, never invent the import
public record Customer(...)  // OK on JDK 21
```

TRY IT NOW Complete Exercise 3 before continuing — see exercises/exercise-03-phantom-annotation-hunt.md.

Generating Service Classes

1 Define Requirements

Identify the service responsibilities, methods, inputs/outputs, dependencies, and business rules.

2 Write a Prompt

Provide a clear and specific prompt describing the service class, methods, parameters, return types, and rules.

3 AI Generates Code

GitHub Copilot analyzes the prompt and generates the service class code.

4 Review & Refine

Review the generated code and refine the prompt or code if changes are needed.

5 Use the Service

Integrate the service class into your application and use it in controllers or other layers.

Example Prompt

Generate a Java Spring service class named UserService with the following:

- Use @Service annotation
- Dependency: UserRepository userRepository
- Methods:
  - UserDto getUserById(Long id)
  - List<UserDto> getAllUsers()
  - UserDto createUser(UserDto userDto)
  - UserDto updateUser(Long id, UserDto userDto)
  - void deleteUser(Long id)
- Use User entity and UserDto
- Add basic business logic and mapping between Entity and DTO

GitHub Copilot

Generated Service Class (UserService.java)

```java
@Service
@RequiedAargsConstructor
public class UserService {
    private final UserRepository userRepository;

    public UserDto getUserById(Long id) {
        User user = userRepository.findById(id)
                .orElseThrow(() -> new ResourceNotFoundException("User not found"));
        return mapToDto(user);
    }

    public List<UserDto> getAllUsers() {
        return userRepository.findAll().stream()
                .map(this::mapToDto)
                .toList();
    }

    public UserDto createUser(UserDto userDto) {
        User user = mapToEntity(userDto);
        User saved = userRepository.save(user);
        return mapToDto(saved);
    }

    public UserDto updateUser(Long id, UserDto userDto) {
        User user = userRepository.findById(id)
                .orElseThrow(() -> new ResourceNotFoundException("User not found"));
        user.setName(userDto.getName());
        user.setEmail(userDto.getEmail());
        user.setActive(userDto.getActive());
        return mapToDto(userRepository.save(user));
    }

    public void deleteUser(Long id) {
        userRepository.deleteById(id);
    }

    private UserDto mapToDto(User user) {
        return new UserDto(user.getId(), user.getName(), user.getEmail(), user.getActive());
    }

    private User mapToEntity(UserDto dto) {
        return new User(null, dto.getName(), dto.getEmail(), dto.getActive());
    }
}
```

GENERATING DOCUMENTATION

Clear documentation makes your code easy to understand, maintain, and use — Copilot can generate it in seconds.

REVIEW CHECKLIST — BEFORE YOU SHIP DOCUMENTATION

- Does the documentation match the code's actual behavior?
- Are the parameter and return descriptions accurate?
- Are exceptions documented?
- Are setup commands verified?
- Does it avoid stating unsupported guarantees?

DOCUMENTATION TOOLS AND TIPS

Copilot can generate several kinds of documentation in seconds — use these tips to keep it accurate.

WHAT COPILOT CAN GENERATE

JavaDoc

Method and class-level docs.

README

Project overview and setup guides.

API Docs

REST endpoint documentation.

Inline Comments

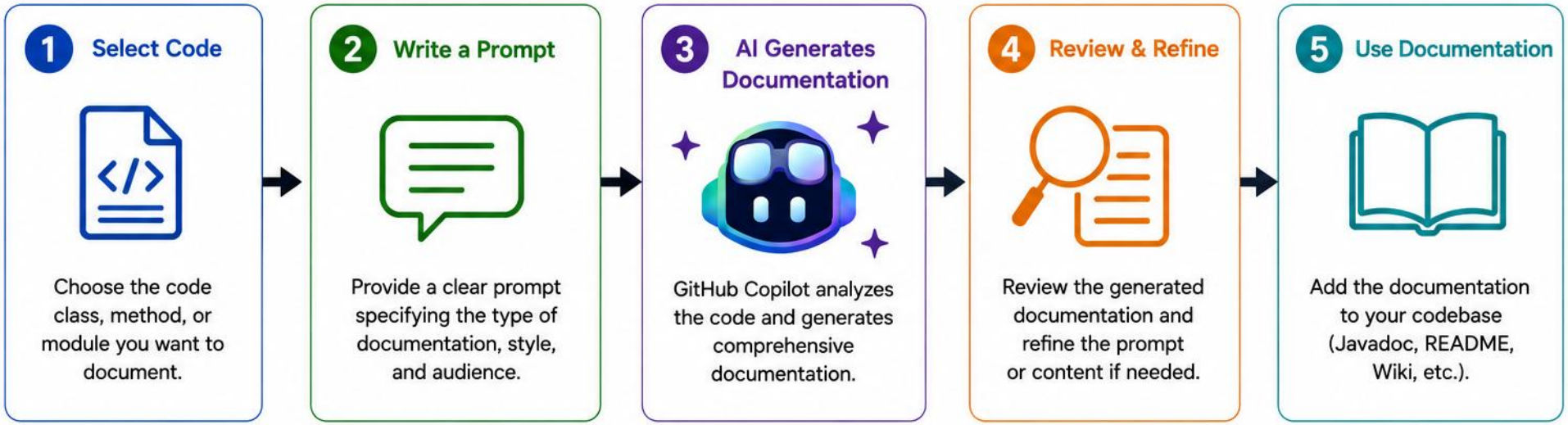
Explain non-obvious logic.

TIPS FOR BETTER DOCUMENTATION

- Be specific in your prompt (e.g. "Generate JavaDoc for this method"). Keep docs in sync with code changes, and follow your project's documentation style and conventions.

KEY TAKEAWAY Good documentation is as important as good code — with Copilot, you can generate high-quality docs quickly.

Generating Documentation



Example Prompt

Generate Javadoc for the following Java method.
Use a professional style and include @param, @return, and @throws tags.

```
public User getUserById(Long id) {
    User user = userRepository.findById(id)
        .orElseThrow(() ->
          new ResourceNotFoundException("User
          not found"));
    return mapToDto(user);
}
```

GitHub Copilot

Generated Documentation

```
/**
 * Retrieves a user by its ID.
 *
 * @param id the unique identifier of the user
 * @return the user DTO if found
 * @throws ResourceNotFoundException if no user
 *         exists with the given ID
 */
public User getUserById(Long id) {
   User user = userRepository.findById(id)
       .orElseThrow(() -> new ResourceNotFoundException("User
         not found"));
   return mapToDto(user);
}
```

REVIEWING AI-GENERATED CODE

AI-generated code can save time, but it's your responsibility to review it for correctness, security, and maintainability. Do: use meaningful names and provide clear, complete prompts. Don't: accept a suggestion blindly or copy code you don't understand.

WHY YOU MUST REVIEW

Correctness
AI can misinterpret intent.

Security
Copy vulnerabilities, hardcoded secrets.

Performance
Can be inefficient or non-optimal.

Maintainability
Should be clean and consistent.

Compliance
Follows your standards and policies.

REVIEW WORKFLOW

1. **Get Suggestion**
2. **Understand Intent**
3. **Review & Analyze**
4. **Refactor & Improve**
5. **Test Thoroughly**
6. **Validate & Accept**

REVIEW IN PRACTICE: A CORRECTNESS BUG

Security isn't the only risk — an unreviewed suggestion can also be logically wrong.

EXAMPLE: OFF-BY-ONE IN A DISCOUNT LOOKUP

```
// UNREVIEWED — Copilot suggestion, misses the last tier
for (int i = 0; i < tiers.size() - 1; i++) {
    if (total.compareTo(tiers.get(i).getThreshold()) >= 0) {
        return tiers.get(i).getDiscount();
    }
}
// REVIEWED & IMPROVED — loop covers every tier, none skipped
for (DiscountTier tier : tiers) {
    if (total.compareTo(tier.getThreshold()) >= 0) {
        return tier.getDiscount();
    }
}
return BigDecimal.ZERO;
```

KEY TAKEAWAY Syntactically correct code can still be logically wrong — trace the loop bounds before you accept.

REVIEW IN PRACTICE: SQL INJECTION

Applying the review workflow to a real suggestion: an unreviewed query is vulnerable until you refactor it.

EXAMPLE: PREPAREDSTATEMENT VS STRING CONCATENATION

```
// UNREVIEWED – SQL injection risk
String sql = "SELECT * FROM user WHERE id=" + id;
// REVIEWED & IMPROVED – parameterized query
String sql = "SELECT * FROM users WHERE id = ?";
try (PreparedStatement stmt = connection.prepareStatement(sql)) {
    stmt.setString(1, id);
}
```

KEY TAKEAWAY Never blindly trust AI-generated code. Always understand before you accept — review smart, deploy with confidence.

TRY IT YOURSELF — EXERCISE 4: FILL REVIEW-LOG TODOS (STARTER)

Reinforces: the six-step review workflow — Get, Understand, Review, Refactor, Test, Validate.

STEP	WHAT TO DO
Goal	FILL-IN-THE-BLANK STARTER — complete a Copilot review log before the timed lab
File	lab10-review-log-todos.md (in examples/module-10-exercises/notes/)
Fill in	6 blanks: prompt strength, phantom annotation, Amina/Ravi fixtures, JDK/Maven note, decision
Rule	Replace every single ____ with a concrete value — never leave a blank empty
Self-check	Confirm Ravi is PROSPECT and Amina is ACTIVE before you claim prep is done
Reference	exercises/exercise-04-fill-review-log-todos.md

```
Prompt strength: _____
Phantom annotation found? _____  // yes/no + name
Fixture check — Amina: _____   Ravi: _____
JDK/Maven note: _____
Accept / Reject / Edit: _____
```

TRY IT NOW Complete Exercise 4 before continuing — see exercises/exercise-04-fill-review-log-todos.md (starter).

Reviewing AI-Generated Code

1 Generate Code

AI generates code based on your prompt and context.

2 Understand the Code

- Read the code carefully
- Understand purpose, inputs, outputs
- Check logic and flow

3 Verify Correctness

- Validate logic
- Check edge cases
- Ensure it meets requirements

4 Check Quality & Best Practices

- Code style & readability
- Naming conventions
- SOLID principles
- Performance
- Error handling
- Security concerns

5 Test Thoroughly

- Run existing tests
- Add missing tests
- Verify all scenarios
- Ensure no regressions

6 Refine & Integrate

- Refactor if needed
- Improve prompts
- Commit & integrate confidently

Code Review Checklist

FUNCTIONALITY

- Does the code meet the requirement?
- Are inputs/outputs handled correctly?
- Are edge cases considered?
- Is the logic correct and complete?

CODE QUALITY

- Is the code readable and well-structured?
- Are names meaningful and consistent?
- Is duplication avoided?
- Is the code easy to maintain?

PERFORMANCE

- Is the algorithm efficient?
- Are loops/queries optimized?
- Is resource usage appropriate?

SECURITY

- Are inputs validated and sanitized?
- Is sensitive data protected?
- Are there any security vulnerabilities?

TESTING

- Are unit tests present and relevant?
- Do tests cover normal, boundary & error cases?
- Do all tests pass?
- Any new tests needed?

DOCUMENTATION

- Is the code documented appropriately?
- Are comments clear and helpful?
- Is the intent clear to future readers?

DECISION

APPROVE
Code is correct, high quality, and ready to integrate.

REFINE
Request changes from AI or refactor for improvement.

REJECT
Code has issues that require major rework.

RISKS OF AI-ASSISTED DEVELOPMENT (1 OF 2)

AI tools boost productivity but introduce new risks — understanding them helps you use AI responsibly.

Incorrect or Inaccurate Code

- Syntactically correct but logically wrong or incomplete.

Security Vulnerabilities

- Insecure patterns or exposed sensitive data.

License & IP Risks

- May reproduce copyrighted code snippets.

Over-Reliance on AI

- Reduced critical thinking and learning.

RISKS OF AI-ASSISTED DEVELOPMENT (2 OF 2)

Four more risks to watch for, plus practical controls for the License & IP Risk category.

Lack of Context Awareness

- Doesn't understand full system or business rules.

Data Privacy Concerns

- Prompts may leak sensitive or proprietary data.

Bias & Ethical Concerns

- May reflect training-data bias, non-inclusive outcomes.

Poor Code Quality

- Can be verbose or misaligned with project standards.

PRACTICAL CONTROLS FOR LICENSE & IP RISK (NOT LEGAL ADVICE)

- Review suggestions for unexpectedly long or distinctive code, and follow your organization's IP policy.
- Use code-referencing/filtering features where available, and validate third-party license compatibility.
- Escalate uncertain cases to legal/compliance.

KEY TAKEAWAY AI is a powerful assistant, not a replacement for engineering judgment. Review, verify, validate — you own the outcome.

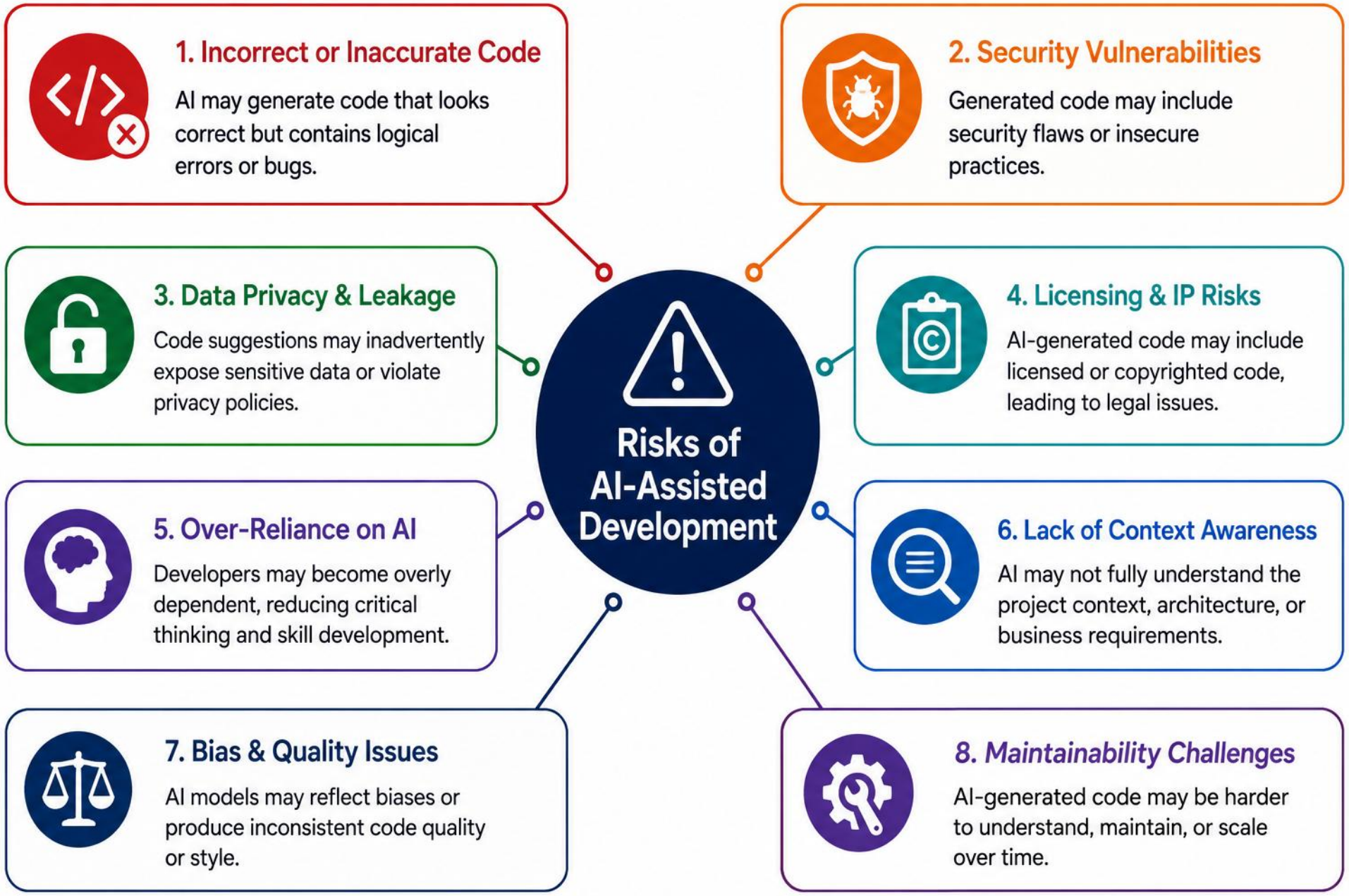
TRY IT YOURSELF — EXERCISE 5: JDK 21 / MAVEN HABIT

Reinforces: Lack of Context Awareness — Copilot doesn't know your real toolchain.

STEP	WHAT TO DO
Goal	Record the exact version checks you will run before Lab 10 coding
File	jdk-maven-checklist.md (in examples/module-10-exercises/notes/)
Commands	java -version (expect 21.x) and mvn -version (expect Java home = 21)
PATH trap	Wrong JDK first on Windows PATH — reorder PATH before the lab, don't assume
Workspace	Bootcamp workspace + notes\ for this module's prep files
Reference	exercises/exercise-05-jdk-maven-habit.md

```
$ java -version
openjdk version "21...  // must read 21.x, not 17 or 11
$ mvn -version
Java version: 21..., Java home: ...jdk-21
# Do not run full lab Maven goals until the timed Lab 10 session
```

TRY IT NOW Complete Exercise 5 before continuing — see exercises/exercise-05-jdk-maven-habit.md.



ENTERPRISE USE CASES

GitHub Copilot supports Java teams across the development lifecycle — from everyday scaffolding to enterprise-scale delivery.

Refactoring Legacy Java

Modernize old modules and update patterns.

Migration Assistance

Help port code between frameworks or versions.

API & Library Exploration

Discover unfamiliar APIs and libraries faster.

Secure Code Review Support

Detect and fix vulnerable patterns.

Test Automation

Unit, integration, and end-to-end tests.

Explaining Unfamiliar Code

Faster ramp-up on unfamiliar codebases.

Documentation

README, API docs, JavaDoc at scale.

Boilerplate & Scaffolding

Generate classes, DTOs, and config fast.

BUSINESS IMPACT AT ENTERPRISE SCALE

These use cases translate into measurable outcomes for engineering teams, across every regulated industry.

Faster
Delivery

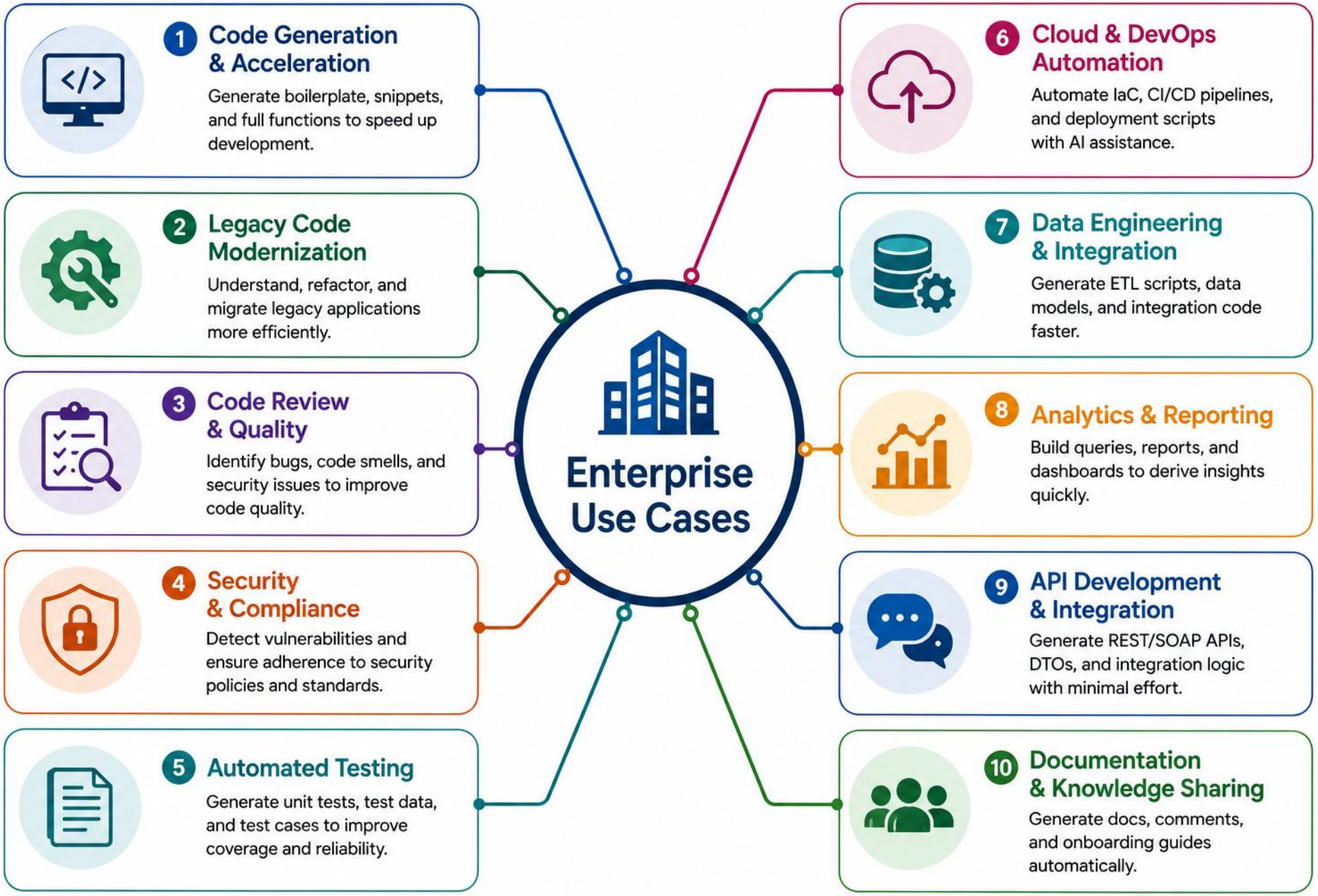
Higher
Quality

Lower
Risk

By Design
Privacy

- Industries: Financial Services, Healthcare, Retail, Manufacturing, Public Sector. Enterprise controls may include organization policies, access management, audit capabilities, content controls, and data-governance options — confirm current capabilities for your GitHub plan.

KEY TAKEAWAY Copilot is more than a coding assistant — it's a force multiplier for engineering teams, used responsibly.



TRY IT YOURSELF — EXERCISE 6: LAB 10 PREP CHECKLIST

Reinforces: Exercises 1–5 together — confirm you're ready before the timed lab.

STEP	WHAT TO DO
Goal	Confirm prep readiness without completing Lab 10 itself
File	lab10-prep-checklist.md (in examples/module-10-exercises/notes/)
Lab asks	Three things from the Lab 10 header: prompts, review log, generated sketch
Fixtures	CUS-1001 Amina ACTIVE, CUS-1002 Ravi PROSPECT, correlation lab-request-001
Pass rule	Pass if prompts + phantom checklist + review TODOs exist; else revisit Exercises 1–4
Reference	exercises/exercise-06-lab10-prep-checklist.md

```
$ cat notes/lab10-prep-checklist.md
[ ] Weak vs strong prompts documented
[ ] Phantom annotations flagged, reject rule written
[ ] Customer sketch: CUS-1001 Amina ACTIVE, CUS-1002 Ravi PROSPECT
[ ] Review-log TODOs filled; java/mvn confirmed on JDK 21
```

TRY IT NOW Complete Exercise 6 before continuing — see exercises/exercise-06-lab10-prep-checklist.md.

LAB OVERVIEW – AI-ASSISTED CODE GENERATION

Use GitHub Copilot to scaffold the Northstar CRM customer domain — with mandatory human review of every suggestion.

LAB SCENARIO

- Extend the Northstar CRM customer-service Maven project (from Lab 9) with a Customer domain.
- Customer has customerId, fullName, email, phone, status, and createdAt.
- Plain Java 21 and Maven only — no Spring, no JPA, no REST yet.
- Required workflow: Prompt → Review → Refine → Test.

LAB ENVIRONMENT

ITEM	VALUE
`Language`	Java 21
`Framework`	None (plain Java + Maven)
`Data Store`	In-memory List<Customer>
`Version Control`	Git & GitHub

REQUIRED WORKFLOW

1 → 2 → 3 → 4

Prompt
Write a clear, specific prompt.

Review
Read and understand the suggestion.

Refine
Adjust the prompt or edit the code.

Test
Verify behavior with the Main harness.

KEY TAKEAWAY Generate, review, refine, and document — use Copilot to accelerate, not to replace your thinking.

LAB TASKS AND DELIVERABLES (1 OF 2)

Complete the tasks, review and refine the AI-generated code, and deliver a working, reviewed customer domain.

#	TASK	DETAILS	WEIGHT
1	Setup & Sign-In	Sign in to Copilot; copy lab9-crm to lab10-crm; verify Copilot status Ready.	10%
2	Weak vs Strong Prompts	Compare vague vs. specific prompts; log entries lab10-001 / lab10-002.	15%
3	Entity, Enum & Service	Scaffold CustomerStatus, Customer, and CustomerService via Copilot Chat.	25%

LAB TASKS AND DELIVERABLES (2 OF 2)

Tasks 4-6, plus the deliverables checklist for what you submit.

#	TASK	DETAILS	WEIGHT
4	Main Harness	Prove behavior for CUS-1001 / CUS-1002; mvn clean compile.	15%
5	Human Review & Experiments	Checklist lab10-003; 3+ documented failure experiments.	15%
6	Risk Notes & Docs	ai-review-notes.md lab10-004; security and production answers.	20%

DELIVERABLES

- Customer entity, CustomerStatus enum, and CustomerService compiling clean; a Main harness proving CUS-1001 / CUS-1002; ai-review-notes.md with entries lab10-001–004; no secrets or target/ committed.
- **Required in ai-review-notes.md:** one accepted suggestion, one modified suggestion, and one rejected suggestion, with the reason for each decision and which tests validated the final code.

KEY TAKEAWAY AI tools can accelerate development, but your judgment, review, and testing deliver quality.

COPILOT DEFINITION OF DONE

Before you call a Copilot-assisted change finished, check it against this practical checklist.

1. Prompt includes the goal, context, constraints, and acceptance criteria.
2. The suggestion is understood — not just accepted.
3. Code follows your project's standards and conventions.
4. Tests cover normal cases and edge cases.
5. Security and privacy have been reviewed.
6. Dependencies and APIs used are verified as real.
7. Documentation matches the code's real behavior.
8. A human reviewer approves the final change.

KEY TAKEAWAY AI suggests, you decide. Review every suggestion, understand before you accept, and build with quality and standards.

KNOWLEDGE CHECK (1 OF 2)

Test your understanding of GitHub Copilot concepts, best practices, and AI-assisted development.

1. What's the primary benefit of GitHub Copilot?

A. It replaces developers
B. It automatically tests the application
C. It writes code faster and reduces busywork
D. It deploys the application to production

2. What should you do right after Copilot generates code?

A. Review and understand the generated code
B. Run it in production immediately
C. Delete it and start over
D. Share it on social media

3. What's a best practice for writing prompts?

A. Be vague to get creative results
B. Provide clear context and specific requirements
C. Ask Copilot to write the entire app in one prompt
D. Avoid describing the expected output

4. What should you always verify in AI-generated code?

A. Variable and method names
B. Syntax and code style
C. Business logic correctness and security
D. All of the above

KEY TAKEAWAY Copilot helps developers write code faster — but correctness, security, and style all still need human review.

KNOWLEDGE CHECK (2 OF 2)

Four more questions, plus the full answer key.

5. What should you do after Copilot generates code?

A. Accept it as-is without any review
B. Run it in production to test
C. Review, understand, test, and refine it
D. Share it with your team immediately

6. Which layer holds business logic in a REST API?

A. Controller Layer
B. Service Layer
C. Repository Layer
D. DTO Layer

7. What's a recommended enterprise Copilot practice?

A. Share all prompts and code publicly
B. Follow organization coding standards
C. Disable code reviews
D. Ignore security scanning

8. What is the main goal of refining AI-generated code?

A. To make it longer
B. To improve readability, performance, and maintainability
C. To change variable names only
D. To avoid writing any code manually

ANSWER KEY

- **1-C 2-A 3-B 4-D 5-C 6-B 7-B 8-B**

KEY TAKEAWAY Use Copilot intentionally. Review critically. Build securely. Document clearly. Deliver high-quality software.

"AI suggests, you decide."

MODULE 10 COMPLETE

GitHub Copilot Fundamentals for Java Developers

You can now use GitHub Copilot to generate, review, and refine code responsibly — a force multiplier, not a replacement for judgment.